MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 1 of 10 (Start time: 03:41:26 PM, 25 June 2025)

$\int_a^b f(x)dx = _____.$

Select the correct option

- $-\int_b^a f(x)dx$
- $\int_b^a f(x)dx$

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 3 of 10 (Start time: 03:42:19 PM, 25 June 2025)

We cannot evaluate definite integrals by substitution method.

Select the correct option

○	True
◉	False

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 2 of 10 (Start time: 03:41:56 PM, 25 June 2025)

Antiderivative of f(x)=x^2 is .....

Select the correct option

○	None of these.
●	(1/3) x^3+c
○	x/2
○	x^3

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 2 of 10 (Start time: 03:48:01 PM, 25 June 2025)

What will be the value of $\int_0^1 e^x dx$?

Select the correct option

○	e^x
●	e
○	1
○	e^{c+1}

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 10 of 10 (Start time: 03:44:50 PM, 25 June 2025)

Antiderivative of f(x) =x is .....

Select the correct option

○	None of these.
◉	(1/2) x^2+c
○	x^3
○	x/2

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 10 of 10 (Start time: 03:51:57 PM, 25 June 2025)

While calculating Riemann sum it is necessary to take equal length subintervals.

Select the correct option

○	True
◉	False

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 6 of 10 (Start time: 03:50:15 PM, 25 June 2025)

The value of the $\int_0^1 t^3 dt =$ _____.

Select the correct option

○	2/3
○	1/3
○	4/3
◉	1/4

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 5 of 10 (Start time: 03:34:05 PM, 25 June 2025)

While calculating Riemann sum it is necessary to take equal length subintervals.

Select the correct option

○	True
◉	False

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 2 of 10 (Start time: 03:33:17 PM, 25 June 2025)

The value of $\int_1^{\infty} \frac{dx}{x^2}$ _____.

Select the correct option

○	3
○	4
◉	1
○	0

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 3 of 10 (Start time: 03:33:32 PM, 25 June 2025)

The integral of f(x) =sin(x+1) from x=0 to x=1 is .....

Select the correct option

○	cos(2)-cos(1)
○	cos(2)-1
◉	cos(1)-cos(2)
○	sin(1)-sin(2)

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 8 of 10 (Start time: 03:51:27 PM, 25 June 2025)

F(x) (antiderivative of f) was determined to be 4x+C where C denotes the .....

Select the correct option

- (●) Integration constant.
- ( ) Differentiation constant.

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 8 of 10 (Start time: 03:28:06 PM, 25 June 2025)

What will be the value of $\int_0^2 (\sin x + 3)dx$ if $\int_0^2 (10\sin x + 30)dx = 74$?

Select the correct option

◉	7.4
○	4.2
○	5.4
○	3.6

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 4 of 10 (Start time: 03:26:27 PM, 25 June 2025)

The value of $\int_1^2 \ln x \, dx$ _____.

Select the correct option

○	2ln2+1
○	ln2-1
○	2ln2+2
◉	2ln2-1

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 10 of 10 (Start time: 03:35:32 PM, 25 June 2025)

$$Evaluate \frac{d}{dx}\int_2^x tdt$$

Select the correct option

○	x^2
○	x^3
◉	x
○	none of these

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 3 of 10 (Start time: 03:26:15 PM, 25 June 2025)

If the integral of $f(x) = x$ from $x = 1$ to $x = 3$ is 4, then the integral of $f(x) = 10x$ from $x = 1$ to $x = 3$ is _____.

Select the correct option

○	4.
○	None of these.
◉	40.
○	20.

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 8 of 10 (Start time: 03:34:55 PM, 25 June 2025)

..... gives a relation between definite integral and indefinite integral.

Select the correct option

●	First fundamental theorem of Calculus
○	Extreme value theorem
○	Mean value theorem
○	None of these.

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 5 of 10 (Start time: 03:26:51 PM, 25 June 2025)

If integral of 'f(x)' from [1,2] = 5 ,and integral of 'f(x)' from [3,2] = 4 , than integral of 'f(x)' from [1,3] is .....

Select the correct option

○	2
○	3
●	1
○	9

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 10 of 10 (Start time: 03:28:39 PM, 25 June 2025)

In order to fully determine the anti derivative of a function f (F(x)), we must have.....

Select the correct option

◉	Integration constant.
○	Boundary conditions
○	None of these
○	Initial conditions.

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 9 of 10 (Start time: 03:51:40 PM, 25 June 2025)

Which of the following is the value of definite integral of 5+5x , where the lower limit is -1 and upper limit is 1 ?

Select the correct option

○	14
○	12
◉	10
○	16

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 9 of 10 (Start time: 03:44:36 PM, 25 June 2025)

If the function and limits of definite integral are the same and variable of integration are changed, i.e

$$\int_a^b f(x)dx = \int_a^b f(t)dt$$

Then the answer would be

Select the correct option

◉	do not changed
○	changed

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 4 of 10 (Start time: 03:33:50 PM, 25 June 2025)

First fundamental theorem of calculus gives the definite integral of a ..... function on a given closed interval in a quick way.

Select the correct option

◉	Continuous
○	Discontinuous

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 4 of 10 (Start time: 03:42:33 PM, 25 June 2025)

If the definite integral of $f(x)=\cos x$ over the interval $[0, a]$ is equal to '1' then what will be the value of a?

Select the correct option

○	2π
●	$\pi/2$
○	π
○	$\pi/4$

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 7 of 10 (Start time: 03:34:32 PM, 25 June 2025)

If the definite integral of $f(x)=\sec^2 x$ over the interval $[0, a]$ is equal to '1' then what will be the value of a?

Select the correct option

○	$\pi/2$
○	π
○	2π
●	$\pi/4$

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 4 of 10 (Start time: 03:49:17 PM, 25 June 2025)

First fundamental theorem of calculus tells us how to evaluate the ..... in a quick way.

Select the correct option

○	Indefinite integral
○	Differential
○	None of these.
●	Definite integral

Question # 7 of 10 (Start time: 03:27:50 PM, 25 June 2025)

If the integral of $f(x) = x$ and $g(x) = 5$ from $x = 2$ to $x = 3$ is 5 / 2 and 5 respectively, then the integral of $h(x) = x + 5$ from $x = 2$ to $x = 3$ is _____.

Select the correct option

○	None of these.
●	15 / 2.
○	7.
○	5.

Question # 3 of 10 (Start time: 03:48:53 PM, 25 June 2025)

The conclusion of the 2nd fundamental theorem of calculus can be expressed in words as: "the derivative of an integral of a function is that....."

Select the correct option

- ○ none of these
- ◉ Original function
- ○ Next function
- ○ Previous function

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 1 of 10 (Start time: 03:47:33 PM, 25 June 2025)

Mathematically second fundamental theorem of calculus can be written as,

Select the correct option

- ○ $\frac{d}{dx}\int_a^t f(t)dt = f(t)$
- ◉ $\frac{d}{dx}\int_a^x f(t)dt = f(x)$
- ○ $\frac{d}{dx}\int_a^t f(t)dt = f'(x)$
- ○ none of these

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 9 of 10 (Start time: 03:28:25 PM, 25 June 2025)

If the integrand is continuous, then the derivative of a definite integral w.r.t its upper limit is equal to the integrand evaluated at the .....

Select the correct option

- ○ middle limit
- ○ None of these
- ○ lower limit
- ◉ upper limit

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 6 of 10 (Start time: 03:27:09 PM, 25 June 2025)

Which of the following statements is true about $\int_0^1 \sec x \tan x dx$?

Select the correct option

- ◉ $\int_0^1 \sec x \tan x dx = [\sec x]_0^1$
- ○ $\int_0^1 \sec x \tan x dx = [\sec x]_0^1 \times [\tan x]_0^1$
- ○ None
- ○ $\int_0^1 \sec x \tan x dx = [\sec x]_0^1 + [\tan x]_0^1$

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 5 of 10 (Start time: 03:49:31 PM, 25 June 2025)

Which of the following statements is true about $[\sin x]_0^2 - [\tan x]_0^2$?

Select the correct option

○	None
●	$[\sin x]_0^2 - [\tan x]_0^2 = [\sin x - \tan x]_0^2$
○	$[\sin x]_0^2 - [\tan x]_0^2 = [2\sin x - 2\tan x]_0^2$
○	$[\sin x]_0^2 - [\tan x]_0^2 = [2\sin x + 2\tan x]_0^2$

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 6 of 10 (Start time: 03:34:17 PM, 25 June 2025)

If the integrand is continuous, then the derivative of a definite integral w.r.t its upper limit is equal to the integrand evaluated at the .....

Select the correct option

○	middle limit
●	upper limit
○	None of these
○	lower limit

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 1 of 10 (Start time: 03:25:26 PM, 25 June 2025)

Mathematically second fundamental theorem of calculus can be written as,

Select the correct option

○	$\frac{d}{dx}\int_a^t f(t)dt = f'(x)$
○	$\frac{d}{dx}\int_a^t f(t)dt = f(t)$
○	none of these
◉	$\frac{d}{dx}\int_a^x f(t)dt = f(x)$

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MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 9 of 10 (Start time: 03:35:16 PM, 25 June 2025)

Which of the following statements is true?

Select the correct option

○	An antiderivative of a sum is the sum of the twice of antiderivatives.
○	A constant factor can be moved through an integral sign only if the constant is positive.
○	An antiderivative of a difference is k-times the difference of the antiderivatives.
◉	None

Question # 1 of 10 (Start time: 03:32:45 PM, 25 June 2025)

Which of the following statements is true?

Select the correct option

- ○ $\int_0^1 (4\sin x + 3\sec^2 x)dx = 3\int_0^1 4\sin x dx + 4\int_0^1 3\sec^2 x dx$
- ○ $\int_0^1 7\cot^3 x dx = \frac{1}{7}\int_0^1 7\cot^3 x dx$
- ◉ $\int_0^1 (\sec x - \tan^2 x)dx = \int_0^1 \sec x dx - \int_0^1 \tan^2 x dx$
- ○ None

Question # 8 of 10 (Start time: 03:44:22 PM, 25 June 2025)

If the integrand is continuous, then the derivative of a definite integral w.r.t its upper limit is equal to the integrand evaluated at the .....

Select the correct option

- ○ middle limit
- ○ lower limit
- ◉ upper limit
- ○ None of these

Click to Save Answer

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Quiz Start Time: 03:25 PM, 25 June 2025

Question # 2 of 10 (Start time: 03:25:46 PM, 25 June 2025)

Total Marks: 1

$$If\ f\ continuous\ on\ [a,b]\ and\ F(x)=\int_a^x f(t)dt,\ then$$

Select the correct option

Reload Math Equations

- $F'(t)=f(x)\ on\ [a,b]$
- $F'(x)=f(t)\ on\ [a,b]$
- (selected) $F'(x)=f(x)\ on\ [a,b]$
- none of these

Click to Save Answer & Move to Next Question

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Quiz Start Time: 03:47 PM, 25 June 2025

Question # 7 of 10 (Start time: 03:50:54 PM, 25 June 2025)

Total Marks: 1

Which of the following statements is true?

Select the correct option

Reload Math Equations

- None
- $\int_0^1 7\cot^3 x dx=\frac{1}{7}\int_0^1 7\cot^3 x dx$
- (selected) $\int_0^1 (2\cos x+6\sin^2 x)dx=\int_0^1 2\cos x dx+\int_0^1 6\sin^2 x dx$
- $\int_0^1 (2\cos x-5\tan^2 x)dx=\int_0^1 5\tan^2 x dx-\int_0^1 2\cos x dx$

Question # 6 of 10 (Start time: 03:43:39 PM, 25 June 2025) **Total Marks:** 1

Which of the following statements is true about $\int_0^1 \sec x \tan x dx$?

Select the correct option

Reload Math Equations

- None
- $\int_0^1 \sec x \tan x dx = [\sec x]_0^1 \times [\tan x]_0^1$
- $\int_0^1 \sec x \tan x dx = [\sec x]_0^1$
- $\int_0^1 \sec x \tan x dx = [\sec x]_0^1 + [\tan x]_0^1$

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No. 4) Quiz Start Time: 03:41 PM, 25 June 2025

Question # 7 of 10 (Start time: 03:44:03 PM, 25 June 2025) **Total Marks:** 1

Which of the following statements is true?

Select the correct option

Reload Math Equations

- $\int_0^1 7\cot^3 x dx = \frac{1}{7}\int_0^1 7\cot^3 x dx$
- $\int_0^1 (\sec x - \tan^2 x)dx = \int_0^1 \sec x dx - \int_0^1 \tan^2 x dx$
- $\int_0^1 (4\sin x + 3\sec^2 x)dx = 3\int_0^1 4\sin x dx + 4\int_0^1 3\sec^2 x dx$
- None

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 03:43:09 PM, 25 June 2025) **Total Marks:** 1

Which of the following statements is true about $\int_0^1 (\cos x + \sec^2 x)dx$?

Select the correct option

Reload Math Equations

- $\int_0^1 (\cos x + \sec^2 x)dx = [\sin x]_0^1 - [\tan x]_0^1$
- None
- $\int_0^1 (\cos x + \sec^2 x)dx = [\sin x]_0^1 \times [\tan x]_0^1$
- $\int_0^1 (\cos x + \sec^2 x)dx = [\sin x]_0^1 + [\tan x]_0^1$

Click to Save Answer & Move to Next Question